

IPA ASSIGNMENT – 5

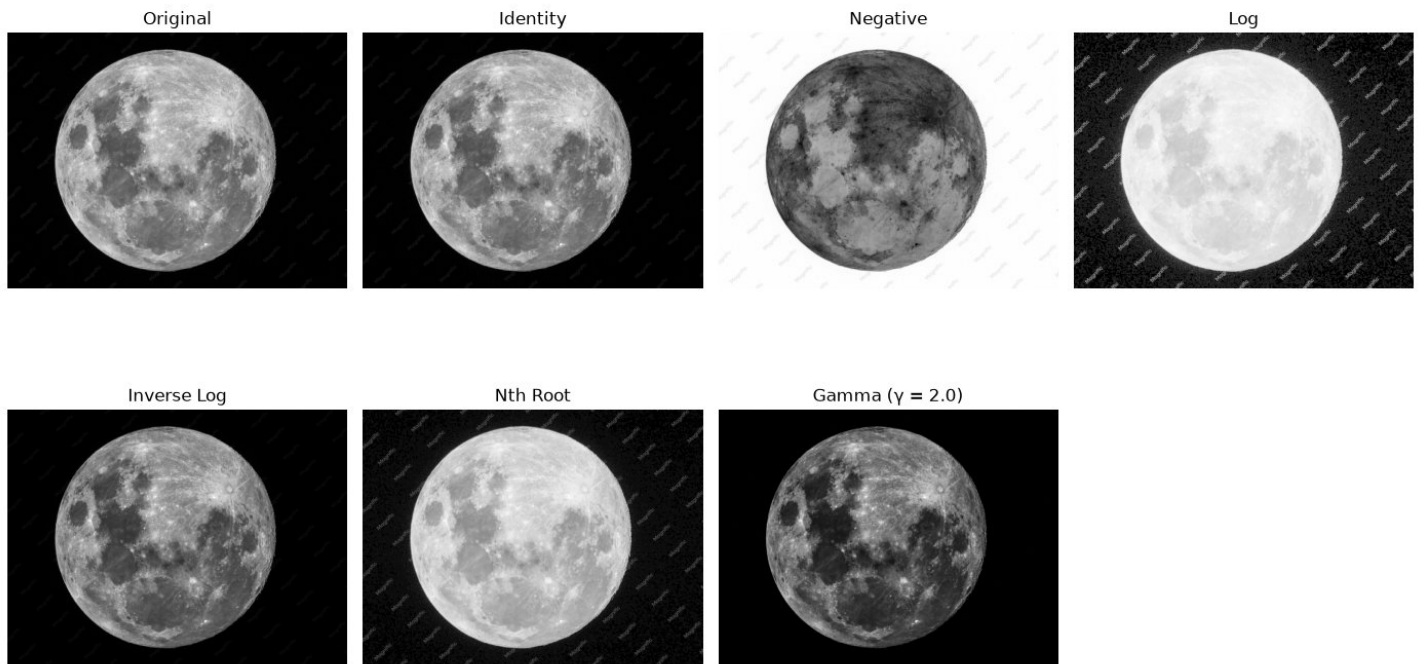
Name – Nishant Nahar

Roll No – 241551078

1. Intensity Transformation – Negative, Identity, Nth Power, Nth Root, Log and Anti Log Transformation

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
img = cv2.imread("image.jpg", cv2.IMREAD_GRAYSCALE)
if img is None:
    raise FileNotFoundError("Image not found!")
img_float = img.astype(np.float32)
img_norm = img_float / 255.0
identity = img.copy()
negative = 255 - img
c = 255 / np.log(1 + np.max(img_float))
log_img = c * np.log(1 + img_float)
log_img = np.clip(log_img, 0, 255).astype(np.uint8)
inv_log = np.exp(img_norm) - 1
inv_log = inv_log / np.max(inv_log)
inv_log = np.clip(inv_log * 255, 0, 255).astype(np.uint8)
n = 2
root_img = np.power(img_norm, 1 / n)
root_img = np.clip(root_img * 255, 0, 255).astype(np.uint8)
gamma = 2.0
gamma_img = np.power(img_norm, gamma)
gamma_img = np.clip(gamma_img * 255, 0, 255).astype(np.uint8)
images = [img, identity, negative, log_img, inv_log, root_img, gamma_img]
titles = [
    "Original",
    "Identity",
    "Negative",
    "Log",
    "Inverse Log",
    "Nth Root",
    "Gamma ( $\gamma = 2.0$ )",
]
plt.figure(figsize=(14, 8))
for i in range(len(images)):
    plt.subplot(2, 4, i + 1)
    plt.imshow(images[i], cmap="gray")
    plt.title(titles[i])
    plt.axis("off")
plt.tight_layout()
plt.show()
```

OUTPUT Images –



2. Gamma Transformation (Gamma = 0.2, 0.5, 1, 2, 5)

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
img = cv2.imread("image.jpg", cv2.IMREAD_GRAYSCALE)
if img is None:
    raise FileNotFoundError("Image not found!")
img_norm = img.astype(np.float32) / 255.0
gamma_values = [0.2, 0.5, 1.0, 2.0, 5.0]
plt.figure(figsize=(15, 8))
plt.subplot(2, 3, 1)
plt.imshow(img, cmap="gray")
plt.title("Original")
plt.axis("off")
for i, gamma in enumerate(gamma_values):
    gamma_img = np.power(img_norm, gamma)
    gamma_img = np.clip(gamma_img * 255, 0, 255).astype(np.uint8)
    plt.subplot(2, 3, i + 2)
    plt.imshow(gamma_img, cmap="gray")
    plt.title(f"Gamma = {gamma}")
    plt.axis("off")
plt.tight_layout()
plt.show()
```

OUTPUT Images

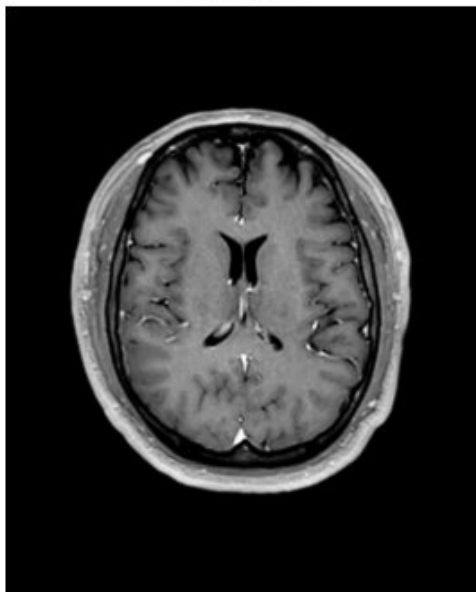


3. Contrast Slicing

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
img = cv2.imread("image_1.jpg", cv2.IMREAD_GRAYSCALE)
if img is None:
    raise FileNotFoundError("Image not found!")
lower = 100
upper = 180
slice_img = img.copy()
slice_img[(img ≥ lower) & (img ≤ upper)] = 255
plt.figure(figsize=(8, 4))
plt.subplot(1, 2, 1)
plt.imshow(img, cmap="gray")
plt.title("Original")
plt.axis("off")
plt.subplot(1, 2, 2)
plt.imshow(slice_img, cmap="gray")
plt.title("Contrast Slicing\n(With Background)")
plt.axis("off")
plt.tight_layout()
plt.show()
```

OUTPUT Images

Original



Contrast Slicing
(With Background)

